Fields and Waves

Exam 1

Fall 2026

1	**/8**
2	**/8**
3	**/8**
4	**/8**
Total	**/32**

Name _____

RIN _____

Notes:
SHOW ALL WORK. BEGIN WITH FORMULAS, THEN SUBSTITUTE VALUES AND UNITS. No credit will be given for answers that appear without justification. Partial credit will be awarded where possible, so show your work so that we can follow your thought process. Where a "Primary Equation" is requested, write down the equation from the crib sheet you used to solve that part of the problem. Use the backs of pages if there is not enough room on the front. If you do, please indicate on the front of your exam that there is more work on the back of pages.

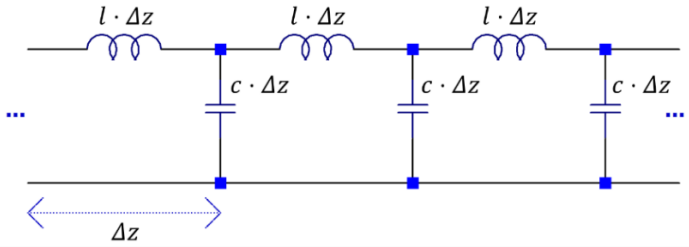
Unless otherwise stated in a problem, provide at least 2 significant digits in answers. Ensure that your calculator is set to use radians for trigonometric functions (and not degrees) and include units with all numerical answers.

At least skim through the entire exam before you begin and then start with the problems you know best.

Problem 1: Transmission Line Fundamentals [8 pts]

a) The transmission line model below consists of segments of length $\Delta z \ll \lambda$ and the following circuit element parameters:

- $l = 0.3125\ \mu\text{H/m} = 3.125 \times 10^{-7}\ \text{H/m}$
- $c = 55.6\ \text{pF/m} = 5.56 \times 10^{-11}\ \text{F/m}$



i) Is the transmission line that is represented by the model above a lossy transmission line or a lossless transmission line? Briefly explain.

The transmission line model represents a lossless line, as it lacks series resistance and parallel conductance elements.

ii) Calculate the velocity factor and characteristic impedance of the transmission line.

Primary Equations:	$\boldsymbol{u = 1/\sqrt{lc}}$ $\boldsymbol{Z_0 = \sqrt{\frac{l}{c}}}$

Velocity Factor:

$$vf = u/c$$

$$vf = \left(\frac{1}{3 \times 10^8 \frac{m}{s}}\right) \frac{1}{\left[\left(3.125 \times 10^{-7} \frac{H}{m}\right)\left(5.56 \times 10^{-11} \frac{F}{m}\right)\right]^{1/2}} = \mathbf{0.8}$$

Characteristic Impedance:

$$Z_0 = \sqrt{\frac{3.125 \times 10^{-7} \frac{H}{m}}{5.56 \times 10^{-11} \frac{F}{m}}} = \mathbf{74.97\Omega}$$

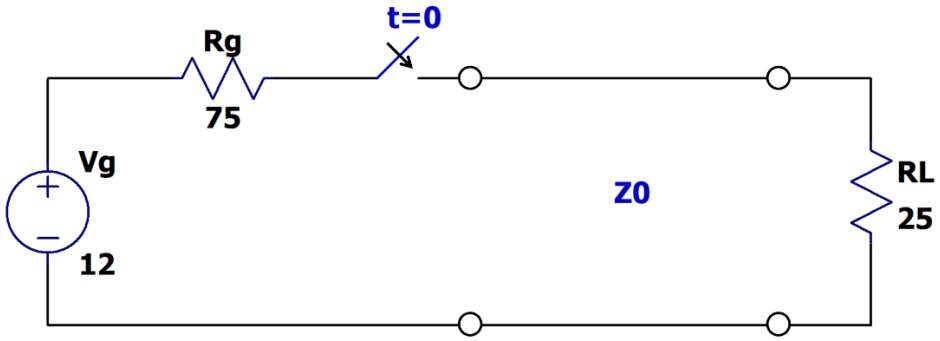
iii) If the transmission line length is $L = 60\text{m}$, calculate the time delay T_d of the transmission line.

Primary Equation:	$L = u \cdot T_d$

$$T_d = L/u = \frac{60\ m}{0.8 \cdot 3 \times 10^8 \frac{m}{s}}$$

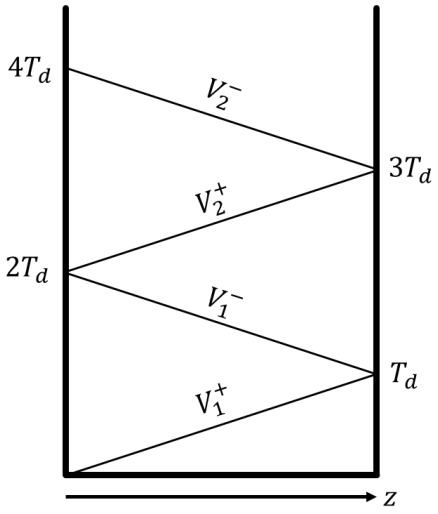
$$T_d = 250 \times 10^{-9}\ s = \mathbf{250\ ns}$$

b) Now consider the DC circuit below, which contains a switch that closes at $t = 0$, and a transmission line with the same properties as specified by the model in part a).



For the bounce diagram below, calculate the following parameters:

- All reflection coefficients
- All voltage amplitudes
- The voltage at the load V_L at $t = T_d$
- The voltage at the input to the transmission line V_{in} at $t = 2T_d$



$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{25\Omega - 75\Omega}{25\Omega + 75\Omega} = \mathbf{-1/2}$$

$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0} = \frac{75\Omega - 75\Omega}{75\Omega + 75\Omega} = \mathbf{0}$$

$$V_1^+ = V_g \cdot \frac{Z_0}{Z_0 + R_g} = 12V \cdot \frac{75\Omega}{75\Omega + 75\Omega} = \mathbf{6V}$$

$$V_1^- = V_1 + \Gamma_L = 6V \cdot \left(-\frac{1}{2}\right) = \mathbf{-3V}$$

$$V_2^+ = V_1^+ \Gamma_L \Gamma_g = \mathbf{0V}$$

$$V_2^- = V_1^+ \Gamma_L^2 \Gamma_g = \mathbf{0V}$$

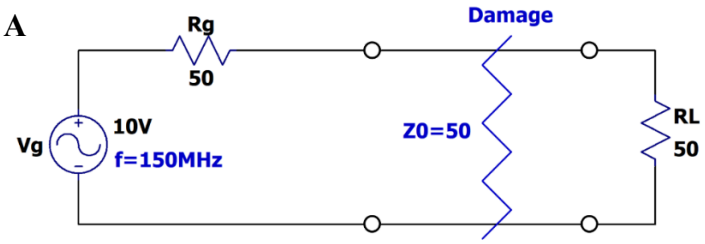
$$V_L(T_d) = V_1^+ V_1^- = 6V + (-3V) = \mathbf{3V}$$

$$V_{in}(2T_d) = V_1^+ + V_1^- + V_2^+ = \mathbf{3V}$$

Problem 2: Damaged Transmission Line [8 pts]

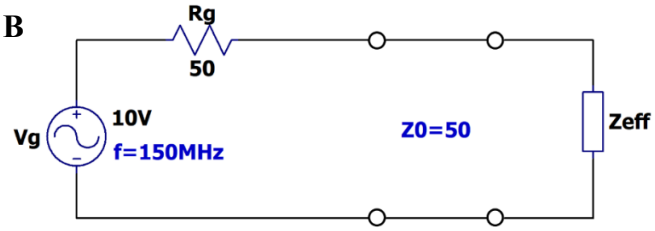
The transmission line in Circuit A below has been damaged, resulting in an abrupt change in its characteristic impedance at a point somewhere along the transmission line.

Circuit A



We can model Circuit A above by replacing the transmission line between the point of the damage and the load by an effective input impedance Z_{eff}, as shown in Circuit B below.

Circuit B



Using a directional coupler, it has been determined that the steady-state forward-traveling voltage wave on the transmission line between its input and Z_{eff} in Circuit B is given by the phasor $\tilde{V}^+(z) = 5e^{-j2\pi z}$ V, and the backward-traveling voltage wave is given by the phasor $\tilde{V}^-(z) = 3e^{j2\pi z}$ V.

a) Write the expression for the total voltage wave on the transmission line between its input and Z_{eff} in the time domain.

$$V(z,t) = Re\left[\left(\tilde{V}^+ + \tilde{V}^-\right)e^{j\omega t}\right]$$
$$= Re\left[\left(5e^{-j2\pi z} + 3e^{j2\pi z}\right)e^{j3\pi\times 10^8\, t}\right] V$$
$$= \mathbf{5\cos(3\pi \times 10^8 t - 2\pi z) + 3\cos(3\pi \times 10^8 t + 2\pi z)\ V}$$

b) Calculate the reflection coefficient between the transmission line and Z_{eff}.

Primary Equation:	$\boldsymbol{\Gamma = \frac{V_0^-}{V_0^+}}$

$$\Gamma_L = \frac{V_0^-}{V_0^+} = \frac{3V}{5V} = \mathbf{0.6}$$

c) Calculate the standing wave ratio.

Primary Equation:	$\boldsymbol{SWR = \frac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}}$

$$SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.6}{1 - 0.6} = \frac{1.6}{0.4} = \mathbf{4}$$

d) Calculate the maximum voltage on the transmission line. How does this voltage compare with the maximum voltage on the transmission line when Circuit A is undamaged? Briefly explain.

Primary Equation:	$\boldsymbol{V_{max} = \lvert V_0^+\rvert + \lvert V_0^-\rvert}$

$$V_{max} = |V_0^+| + |V_0^-| = 5V + 3V = \mathbf{8V}$$

The maximum voltage in this case is larger than when Circuit A is undamaged. In Circuit A, the line is matched to the load and there are no reflections, so $V_{max} = 5V$.

e) Calculate the time-average power delivered to Z_{eff}. How does this delivered power compare with the power delivered to the load when Circuit A is undamaged? Briefly explain.

Primary Equation:	$\boldsymbol{P_{ave} = \frac{\lvert V_0^+\rvert^2}{2Z_0}\left(1 - \lvert\Gamma\rvert^2\right)}$

$$V_{max} = P_{ave} = \frac{|V_0^+|^2}{2Z_0}(1 - |\Gamma|^2) = \frac{|5V|^2}{2 \cdot 50\Omega}(1 - 0.6^2) = \mathbf{0.16W}$$

The delivered power in this case is lower than when Circuit A is undamaged. Since the line and load are matched in Circuit A, $P_{avg} = 0.25W$, since there is no reflected power.

f) Calculate the value of the impedance Z_{eff}. Did the damage to the transmission line increase or decrease its characteristic impedance at that location?

Primary Equation:	$\boldsymbol{\Gamma = \frac{Z_{eff} - Z_0}{Z_{eff} + Z_0}}$

$$\Gamma = \frac{Z_{eff} - Z_0}{Z_{eff} + Z_0} \rightarrow \Gamma(Z_{eff} + Z_0) = Z_{eff} - Z_0$$
$$Z_{eff}(\Gamma - 1) = -Z_0(\Gamma + 1)$$
$$Z_{eff} = -Z_0\frac{\Gamma + 1}{\Gamma - 1} = -50\Omega\frac{0.6 + 1}{0.6 - 1} = \mathbf{200\Omega}$$

The damage increased Z_0.

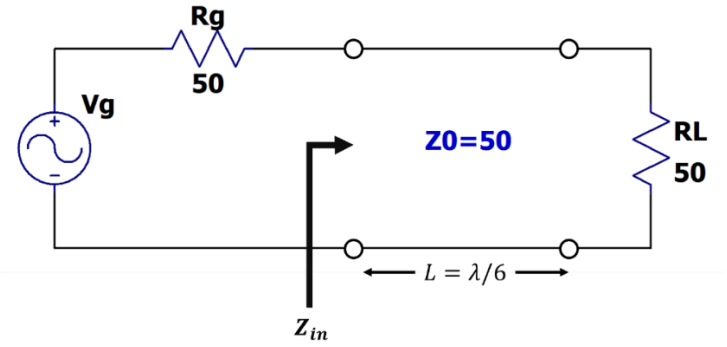
Problem 3: Impedance Matching [8 pts]

In each of the cases below:

1. Calculate Z_{in}.
2. State whether or not the generator circuit sees a matched load.

Briefly justify your answer using a transmission line property or impedance matching method. Answers relying solely on the general mathematical equation for Z_{in} will receive reduced credit.

a) The generator circuit is connected to the load via a transmission line of length $L = \lambda/6$ and characteristic impedance $Z_0 = 50\Omega$.

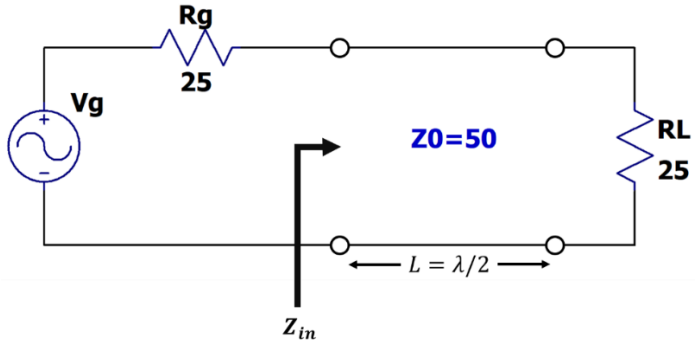


$Z_{\text{in}} = \mathbf{50\Omega}$

Since $R_L = 50\Omega = Z_0$, the transmission line is matched to the load.

Matched? **Yes**

b) The generator circuit is connected to the load via a transmission line of length $L = \lambda/2$ and characteristic impedance $Z_0 = 50\Omega$.

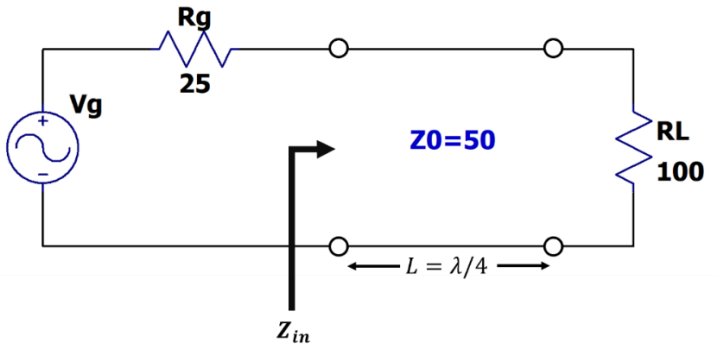


$Z_{\text{in}} = \mathbf{25\Omega}$

For $L = \lambda/2$, $Z_{in} = R_L$, so $Z_{in} = 25\Omega$, which is matched to R_g.

Matched? **Yes**

c) The generator circuit is connected to the load via a transmission line of length $L = \lambda/4$ and characteristic impedance $Z_0 = 50\Omega$.



$Z_{\text{in}} = \mathbf{25\Omega}$

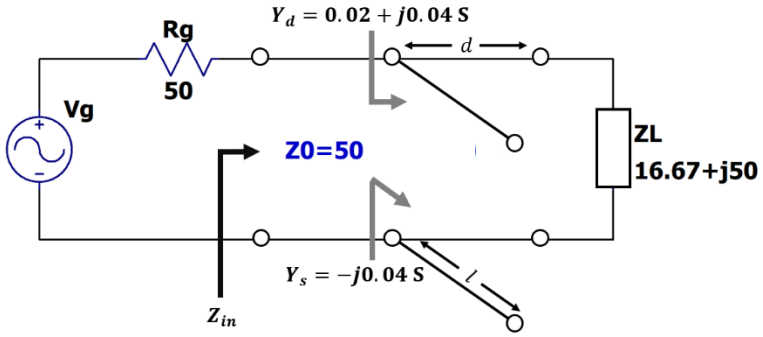
For $L = \lambda/4$, $Z_{in} = Z_0^2/R_L \rightarrow Z_{in} = \frac{(50\Omega)^2}{100\Omega} = 25\Omega$

So, the transmission line + load are matched to the generator.

Matched? **Yes**

d) The generator circuit is connected to a feedline of characteristic impedance $Z_0 = 50\Omega$. The feedline is connected to the parallel combination of two other transmission lines:

- a line of length d, input admittance Y_d, and terminating impedance Z_L
- a line of length l, input admittance Y_s, and an open-circuit termination



$Z_{\text{in}} = \mathbf{50\Omega}$

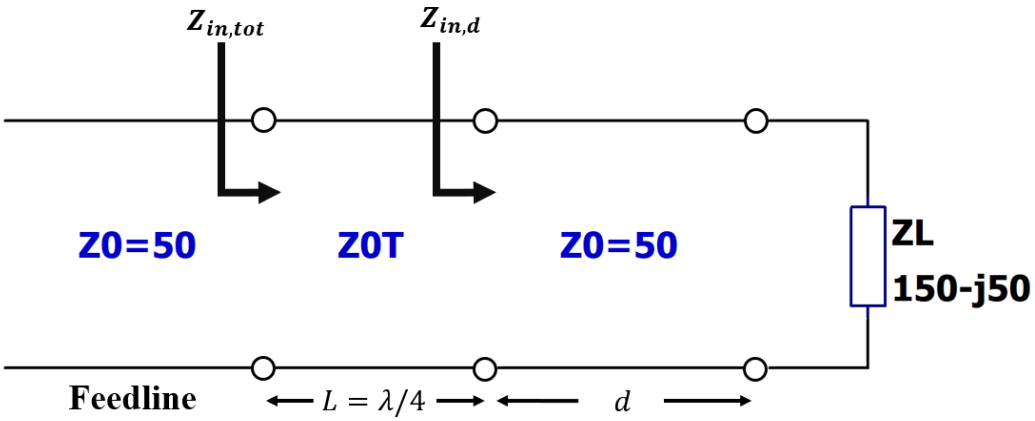
$$Z_{in} = \frac{1}{Y_{in}} = \frac{1}{Yd + Ys} = \frac{1}{(0.02 + j0.04)S - j0.04S}$$
$$= \frac{1}{0.02S} = 50\Omega$$

The transmission line network is matched to the generator via single stub matching.

Matched? **Yes**

Problem 4: Impedance Matching with a Smith Chart [8 pts]

The load in the circuit below is to be matched to a $Z_0 = 50\Omega$ feedline using a quarter-wave transformer of characteristic impedance $Z0T$ and a length of transmission line d of characteristic impedance Z_0. The velocity factor of the transmission line of length d is 0.85 and the frequency of the signal on the line is $f = 100\ \text{MHz} = 100 \times 10^6$ Hz.



a) Calculate the normalized load impedance. Plot it on the Smith chart.

Primary Equation:	$\mathbf{z_L = Z_L/Z_0}$

$$z_L = Z_L/Z_0 = \frac{(150 - j50)\Omega}{50\Omega} = \mathbf{3 - j}$$

b) What are the two values of purely real input impedance $Z_{in,d}$ that the transmission line of length d could present? Show your work on the Smith chart. Give your impedances in unnormalized format.

$Z_{in,d,1} = \mathbf{14.5\Omega}$

$$Z_{in,d,1} = 0.26 \cdot 50\Omega = 14.5\Omega$$

$Z_{in,d,2} = \mathbf{170\Omega}$

$$Z_{in,d,2} = 3.4 \cdot 50\Omega = 170\Omega$$

c) Choose one of the two values of $Z_{in,d}$ from part b). How long d (in meters) must the transmission line be to present your chosen input impedance $Z_{in,d}$? Show your work on the Smith chart.

$$\lambda = \frac{u}{f} = \frac{\left(0.85 \cdot 3 \times 10^8 \frac{m}{s}\right)}{100 \times 10^6 Hz} = 2.55\ m$$

For $Z_{in,d} = 14.5\Omega$: $d = 0.5\lambda - 0.267\lambda = 0.233\lambda \rightarrow d = \mathbf{0.594\ m}$

OR

For $Z_{in,d} = 170\Omega$: $d = 0.233\lambda + 0.25\lambda = 0.483\lambda \rightarrow d = \mathbf{1.23\ m}$

d) For the case you chose in part c), find the characteristic impedance Z_{0T} of the quarter-wave transformer that matches the load network to the $Z_0 = 50\Omega$ feedline.

Primary Equation:	$\boldsymbol{Z_{in} = Z_{0T}^2 / Z_L}$

For $Z_{in,d} = 14.5\Omega$: $Z_{0T} = \sqrt{50\Omega \cdot 14.5\Omega} = \mathbf{26.93\ \Omega}$

OR

For $Z_{in,d} = 170\Omega$: $Z_{0T} = \sqrt{50\Omega \cdot 170\Omega} = \mathbf{92.20\ \Omega}$

e) Once the transmission line and load are matched to the feedline, maximum power will be transferred from the generator to the load in steady state. Where do reflections still occur in the system? Circle all that apply.

a. At the interface between the feedline and the input to the $L = \lambda/4$ line.

b. At the interface between the $L = \lambda/4$ line and the input to the line of length d.

c. At the interface between the line of length d and the load Z_L.

d. No reflections occur anywhere in the system since the feedline is matched to the load.

The Complete Smith Chart

Black Magic Design

